Администрирование сетевых подсистем

Лабораторная работа №10 — Расширенные настройки SMTP-сервера

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08 января 2026

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Цели и задачи работы

Цель лабораторной работы

Приобрести практические навыки конфигурирования SMTP-сервера в части настройки аутентификации и защищённой передачи данных.

Задачи

- Настроить доставку почты через LMTP (Postfix → Dovecot)
- Включить SMTP-аутентификацию (SASL через Dovecot)
- Настроить SMTP over TLS (STARTTLS) и порт submission 587
- Проверить работу через консольные тесты и почтовый клиент
- Зафиксировать изменения в provisioning-окружении Vagrant

Выполнение лабораторной работы

Добавление LMTP в список протоколов

Open | dovecot.conf /etc/dovecot | Save

```
17 # those. These are exceptions to this though: No sections (e.g. namespace {})
18 # or plugin settings are added by default, they're listed only as examples.
19 # Paths are also just examples with the real defaults being based on configure
20 # options. The paths listed here are for configure --prefix=/usr
21 # --sysconfdir=/etc --localstatedir=/var
22
23 # Protocols we want to be serving.
24
25 protocols = imap pop3 lmtp
26
27 # A comma separated list of IPs or hosts where to listen in for connections.
28 # "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
29 # If you want to specify non-default ports or anything more complex,
30 # edit conf.d/master.conf.
31 #listen = *, ::
32
33 # Base directory where to store runtime data.
34 #base_dir = /var/run/dovecot/
35
36 # Name of this instance. In multi-instance setup doveadm and other commands
37 # can use -i <instance_name> to select which instance is used (an alternative
38 # to -c <config_path>). The instance name is also added to Dovecot processes
39 # in ps output.
40 #instance_name = dovecot
```

Настройка сервиса LMTP и unix-сокета

Open

10-master.conf
/etc/dovecot/conf.d

```
52   inet_listener submissions {
53     #port = 465
54   }
55 }
56
57 service lmtp
58   unix_listener /var/spool/postfix/private/dovecot-lmtp {
59     group = postfix
60     user = postfix
61     mode = 0600
62   }
63
64
65
66 service imap {
```

Рис. 2: Сервис LMTP в 10-master.conf

Формат имени пользователя для аутентификации

Open | 10-auth.conf /etc/dovecot/conf.d | Save

```
40 #auth_username_chars = abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ01234567890.-_@
41
42 # Username character translations before it's looked up from databases. The
43 # value contains series of from -> to characters. For example "#@/@" means
44 # that '#' and '/' characters are translated to '@'.
45 #auth_username_translation =
46
47 # Username formatting before it's looked up from databases. You can use
48 # the standard variables here, eg. %Lu would lowercase the username, %n would
49 # drop away the domain if it was given, or "%n-AT-%d" would change the '@' into
50 # "-AT-". This translation is done after auth_username_translation changes.
51
52 auth_username_format = %Ln
53
54 # If you want to allow master users to log in by specifying the master
55 # username within the normal username string (ie. not using SASL mechanism's
56 # support for it), you can specify the separator character here. The format
57 # is then <username><separator><master username>. UW-IMAP uses "*" as the
58 # separator, so that could be a good choice.
59 #auth_master_user_separator =
60
```

Рис. 3: auth_username_format в 10-auth.conf

Доставка сообщения: анализ логов

```
Jan  8 09:55:37 server postfix/smtpd[23665]: connect from client.akabrikosov.net[192.168.1.30]
Jan  8 09:55:37 server postfix/smtpd[23665]: 46B3E412C6E9: client=client.akabrikosov.net[192.168.1.30
]
Jan  8 09:55:37 server postfix/cleanup[23669]: 46B3E412C6E9: message-id=<20260108095537.3CCFA6132098@
client.akabrikosov.net>
Jan  8 09:55:37 server postfix/smtpd[23665]: disconnect from client.akabrikosov.net[192.168.1.30] ehl
o=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Jan  8 09:55:37 server postfix/qmgr[23573]: 46B3E412C6E9: from=<akabrikosov@client.akabrikosov.net>,
size=576, nrcpt=1 (queue active)
Jan  8 09:55:37 server postfix/local[23670]: 46B3E412C6E9: passing <akabrikosov@akabrikosov.net> to t
ransport=lmtp
Jan  8 09:55:37 server dovecot[23612]: lmtp(23672): Connect from local
Jan  8 09:55:37 server dovecot[23612]: lmtp(akabrikosov)<23672><Bh8IEhl/X2l4XAAA7sVfPw>: msgid=<20260
108095537.3CCFA6132098@client.akabrikosov.net>: saved mail to INBOX
Jan  8 09:55:37 server dovecot[23612]: lmtp(23672): Disconnect from local: Logged out (state=READY)
Jan  8 09:55:37 server postfix/lmtp[23671]: 46B3E412C6E9: to=<akabrikosov@akabrikosov.net>, relay=ser
ver.akabrikosov.net[private/dovecot-lmtp], delay=0.02, delays=0/0/0.01/0.01, dsn=2.0.0, status=sent (
250 2.0.0 <akabrikosov@akabrikosov.net> Bh8IEhl/X2l4XAAA7sVfPw Saved)
Jan  8 09:55:37 server postfix/qmgr[23573]: 46B3E412C6E9: removed
```

Рис. 4: Логи доставки через LMTP

Проверка почтового ящика Maildir

```
[akabrikosov@server.akabrikosov.net ~]$ MAIL=~/Maildir/ mail
s-nail version v14.9.24.  Type `?' for help
/home/akabrikosov/Maildir: 3 messages 1 new 2 unread
    1 akabrikosov           2026-01-08 09:38   18/685   "test1
 U  2 akabrikosov           2026-01-08 09:40   18/685   "test3
►N  3 akabrikosov@client.a  2026-01-08 09:55   21/880   "LMTP test
& q
Held 3 messages in /home/akabrikosov/Maildir
[akabrikosov@server.akabrikosov.net ~]$
```

Рис. 5: Письмо доставлено в Maildir

Служба auth в Dovecot

Open

10-master.conf
/etc/dovecot/conf.d

```
84
85 service auth {
86   unix_listener auth-userdb {
87     mode = 0600
88     user = dovecot
89   }
90
91   unix_listener /var/spool/postfix/private/auth {
92     group = postfix
93     user = postfix
94     mode = 0660
95   }
96
97 }
98
```

Рис. 6: service auth в 10-master.conf

Настройка SASL в Postfix

```
[root@server.akabrikosov.net server]# gedit /etc/dovecot/conf.d/10-master.conf
[root@server.akabrikosov.net server]#
[root@server.akabrikosov.net server]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server.akabrikosov.net server]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server.akabrikosov.net server]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_reci
pient_domain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_unverif
ied_recipient, permit'
[root@server.akabrikosov.net server]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server.akabrikosov.net server]#
```

Рис. 7: Postfix: параметры SASL

Временная конфигурация для теста SMTP AUTH

```
13 #smtp      inet  n       -       n       -       1       postscreen
14 #smtpd     pass  -       -       n       -       -       smtpd
15 #dnsblog   unix  -       -       n       -       0       dnsblog
16 #tlsproxy  unix  -       -       n       -       0       tlsproxy
17 # Choose one: enable submission for loopback clients only, or for any client.
18 #127.0.0.1:submission inet n -   n       -       -       smtpd
19 #submission inet n       -       n       -       -       smtpd
20 #  -o syslog_name=postfix/submission
21 #  -o smtpd_tls_security_level=encrypt
22   -o smtpd_sasl_auth_enable=yes
23 #  -o smtpd_tls_auth_only=yes
24 #  -o local_header_rewrite_clients=static:all
25 #  -o smtpd_reject_unlisted_recipient=no
26 #     Instead of specifying complex smtpd_<xxx>_restrictions here,
27 #     specify "smtpd_<xxx>_restrictions=$mua_<xxx>_restrictions"
28 #     here, and specify mua_<xxx>_restrictions in main.cf (where
29 #     "<xxx>" is "client", "helo", "sender", "relay", or "recipient").
30 #  -o smtpd_client_restrictions=
31 #  -o smtpd_helo_restrictions=
32 #  -o smtpd_sender_restrictions=
33 #  -o smtpd_relay_restrictions=
34   -o
   smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown_recipient_domain,permit_sasl
35 #  -o milter_macro_daemon_name=ORIGINATING
```

Рис. 8: master.cf для тестирования SMTP AUTH

Проверка SMTP AUTH

```
EHLO test
250-server.akabrikosov.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN YWthYnJpa29zb3YAYWthYnJpa29zb3YAMTIzNDU2
235 2.7.0 Authentication successful
quit
221 2.0.0 Bye
Connection closed by foreign host.
[root@client.akabrikosov.net ~]#
```

Подключение TLS (сертификат Dovecot)

```
[root@server.akabrikosov.net server]#
[root@server.akabrikosov.net server]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs/
[root@server.akabrikosov.net server]# cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private/
[root@server.akabrikosov.net server]# postconf -e 'smtpd_tls_cert_file = /etc/pki/tls/certs/dovecot.p
em'
[root@server.akabrikosov.net server]# postconf -e 'smtpd_tls_key_file = /etc/pki/tls/private/dovecot.
pem'
[root@server.akabrikosov.net server]# postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/
postfix/smtpd_scache'
[root@server.akabrikosov.net server]# postconf -e 'smtpd_tls_security_level = may'
[root@server.akabrikosov.net server]# postconf -e 'smtp_tls_security_level = may'
[root@server.akabrikosov.net server]#
```

Рис. 10: TLS-параметры Postfix

Порт submission (587) и обязательный TLS

Open | master.cf /etc/postfix | Save

```
14 #smtp      pass  -       -       n       -       -       smtpd
15 #dnsblog   unix  -       -       n       -       0       dnsblog
16 #tlsproxy  unix  -       -       n       -       0       tlsproxy
17 # Choose one: enable submission for loopback clients only, or for any client.
18 #127.0.0.1:submission inet n -   n       -       -       smtpd
19 submission inet n       -       n       -       -       smtpd
20 #  -o syslog_name=postfix/submission
21   -o smtpd_tls_security_level=encrypt
22   -o smtpd_sasl_auth_enable=yes
23 #  -o smtpd_tls_auth_only=yes
24 #  -o local_header_rewrite_clients=static:all
25 #  -o smtpd_reject_unlisted_recipient=no
26 #     Instead of specifying complex smtpd_<xxx>_restrictions here,
27 #     specify "smtpd_<xxx>_restrictions=$mua_<xxx>_restrictions"
28 #     here, and specify mua_<xxx>_restrictions in main.cf (where
29 #     "<xxx>" is "client", "helo", "sender", "relay", or "recipient").
30 #  -o smtpd_client_restrictions=
31 #  -o smtpd_helo_restrictions=
32 #  -o smtpd_sender_restrictions=
33 #  -o smtpd_relay_restrictions=
34  -o
   smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown_recipient_domain,permit_sasl_
35 #  -o milter_macro_daemon_name=ORIGINATING
36 # Choose one: enable submissions for loopback clients only, or for any client.
```

Рис. 11: Сервис submission в master.cf

Проверка STARTTLS и аутентификации на 587

```
read R BLOCK
EHLO test
250-server.akabrikosov.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN YWthYnJpa29zb3YAYWthYnJpa29zb3YAMTIzNDU2
235 2.7.0 Authentication successful
quit
221 2.0.0 Bye
closed
[root@client.akabrikosov.net ~]#
```


Рис. 12: Проверка SMTP over TLS

Настройка Evolution

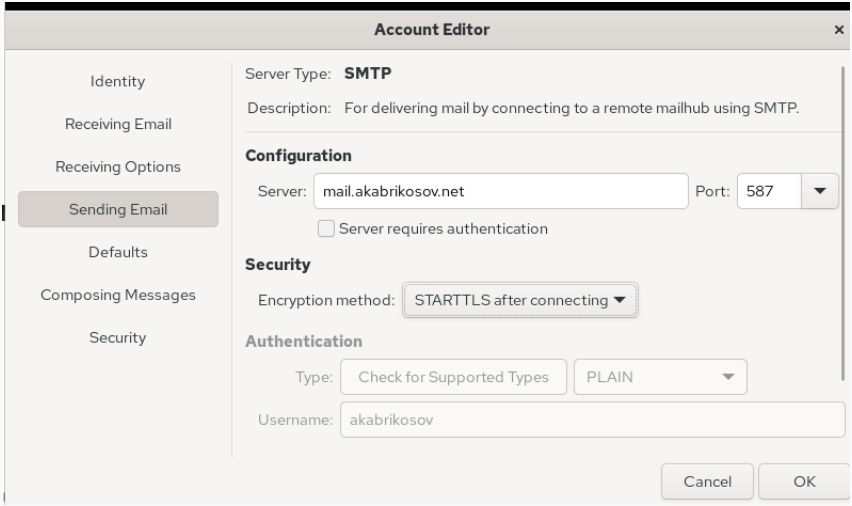

Рис. 13: Параметры SMTP в Evolution

Результат проверки

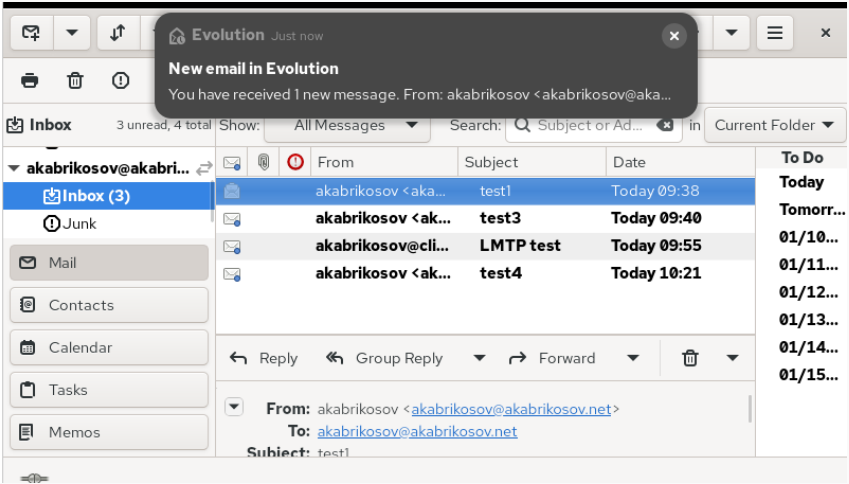

Рис. 14: Сообщения в Evolution

Логи сервера при работе TLS и AUTH

```
Jan  8 10:21:23 server postfix/smtpd[27242]: 152C8412AD5C: client=client.akabrikosov.net[192.168.1.30
], sasl_method=PLAIN, sasl_username=akabrikosov
Jan  8 10:21:23 server postfix/cleanup[27381]: 152C8412AD5C: message-id=<1912a828556e34c931e40de8aa62
edaac68ded2f.camel@akabrikosov.net>
Jan  8 10:21:23 server postfix/qmgr[27207]: 152C8412AD5C: from=<akabrikosov@akabrikosov.net>, size=57
2, nrcpt=1 (queue active)
Jan  8 10:21:23 server postfix/smtpd[27242]: disconnect from client.akabrikosov.net[192.168.1.30] ehl
o=2 starttls=1 auth=1 mail=1 rcpt=1 data=1 rset=1 quit=1 commands=9
Jan  8 10:21:23 server postfix/local[27382]: 152C8412AD5C: passing <akabrikosov@akabrikosov.net> to t
ransport=lmtp
Jan  8 10:21:23 server dovecot[25069]: lmtp(27384): Connect from local
Jan  8 10:21:23 server dovecot[25069]: lmtp(akabrikosov)<27384><kGftBSOFX2n4agAA7sVfPw>: msgid=<1912a
828556e34c931e40de8aa62edaac68ded2f.camel@akabrikosov.net>: saved mail to INBOX
Jan  8 10:21:23 server postfix/lmtp[27383]: 152C8412AD5C: to=<akabrikosov@akabrikosov.net>, relay=ser
ver.akabrikosov.net[private/dovecot-lmtp], delay=0.02, delays=0.01/0/0.01/0, dsn=2.0.0, status=sent (
250 2.0.0 <akabrikosov@akabrikosov.net> kGftBSOFX2n4agAA7sVfPw Saved)
Jan  8 10:21:23 server dovecot[25069]: lmtp(27384): Disconnect from local: Logged out (state=READY)
Jan  8 10:21:23 server postfix/qmgr[27207]: 152C8412AD5C: removed
Jan  8 10:21:26 server dovecot[25069]: imap-login: Login: user=<akabrikosov>, method=PLAIN, rip=192.1
68.1.30, lip=192.168.1.1, mpid=27399, TLS, session=<lfh2xd1HcITAqAEe>
```

Рис. 15: Фрагмент maillog: TLS + SASL + LMTP

Итоги работы

Вывод

- Настроена доставка почты через LMTP (Postfix → Dovecot)
- Реализована SMTP-аутентификация пользователей (SASL через Dovecot)
- Включён SMTP over TLS и настроен submission 587 с обязательным шифрованием
- Выполнено тестирование через консоль и Evolution
- Конфигурация перенесена в provisioning, обеспечено воспроизводимое развёртывание